Project proposal

Base Position as a Redundancy Parameter: Branch-Resolved IK, Workspace Structure and Path Planning for a Mobile Base + UR12e

Motivation

An AgileX UMR omnidirectional base carries a 6R UR12e arm performing a full six-dimensional Cartesian task. Taken as one system the robot is redundant: a single tool pose is realised by a continuous family of configurations rather than a discrete set. For this project the base is restricted to translation along the world X axis and treated as a prismatic joint, giving a P6R robot with seven degrees of freedom and a one-dimensional self-motion for every reachable pose.

Fixing the base position $x = a$ reduces the robot to a UR12e whose target has been shifted by $-a$ along X. Every value of a therefore produces the same finite set of IK branches, applied to a different target. The redundancy becomes visible only when x varies while the task stays fixed: the base position then decides which branches exist, how far each is from a joint limit or a singularity, and whether a continuous execution is possible.

The central question of this project is:

> **How does the base position x select among the IK branches and reachable regions of the UR12e, and how can x and the branch be planned jointly so that a Cartesian task is executed continuously?**

Task

Kinematic model.

No course asset exists for this robot. Assemble a URDF of the whole body from the Universal Robots description package (UR12e; UR10e as fallback) and the UMR mounting geometry, with the base translation as a prismatic joint along world X. Identify the seven joint axes, fixed offsets, joint limits, base frame and flange frame. The physical UR joints span $\pm 2\pi$; for this project the limits are taken as $\pm\pi$ on every joint, so after wrapping each geometric branch yields exactly one candidate joint vector. Build a consistent DH model and the seven screw axes in a defined home configuration. Derive the forward kinematics by DH and by Product of Exponentials and show numerically that both agree with the URDF.

Jacobians.

From the screws construct the complete 6×7 geometric Jacobian, the reduced 6×6 Jacobian of the arm for fixed x, and the wrist/subchain Jacobian ${}^{3}J_{5}$ used in the course, interpreting the columns geometrically. Use the full and reduced Jacobians to separate a true singularity of the redundant robot from a singularity of the fixed-x representation. Since the prismatic column is a pure X translation, the full Jacobian loses rank only when the reduced one is singular and the lost direction has no X component: a wrist singularity should remain a whole-body singularity, while shoulder and elbow singularities should be partly recoverable by the base. Verify this numerically.

Analytical IK solver.

For fixed x, derive a complete analytical IK that exposes all regular discrete solution branches of the resulting 6R problem. The UR wrist is offset, so follow the UR-specific closed form rather than a Pieper decomposition. The implementation must return all real solutions, wrap every angle to $(-\pi, \pi]$, apply the $\pm\pi$ joint limits, and verify every retained solution through the independent forward model. Label each solution by shoulder, elbow and wrist choice (up to eight branches) and check that labels persist between neighbouring poses. Validate by round trip from random joint vectors, and vectorise over arrays of poses since the maps below call the solver millions of times.

Reachable-position maps.

With the IK available, fix the flange orientation (tool down and tool horizontal) and sweep the flange position over a 2D world-frame grid. Record per branch one of four outcomes: feasible, no real IK, joint-limit rejection, or loss of regularity (smallest singular value of the reduced Jacobian below a threshold). Repeat for $x = -1, 0, +1$ m and overlay the maps in the world frame; since fixing x only translates the arm, the content lies in the overlaps and gaps. Relate the region boundaries to the reduced and full Jacobians (critical curves) rather than treating the workspace as a numerical point cloud.

Cartesian path and the (s, x) redundancy plane.

Select a pick-and-place path, parametrised by $s \in [0, 1]$, whose span exceeds the arm's comfortable reach and crosses an informative region of the maps. (1) **Fixed-x baseline:** at $x = 0$ m start from every admissible IK solution and follow each branch separately, recording where it completes, loses IK, hits a limit, loses continuity or drops below the singularity margin; repeat for one other x. (2) **Redundancy plane:** for each branch evaluate the four-class outcome over a grid in (s, x). A fixed-x execution is a horizontal line; a moving-base execution is any continuous curve through a feasible region, with the base speed limit bounding dx/ds. (3) **Planning comparison:** compare greedy continuation (nearest configuration, x chosen locally) with a branch-aware planner that keeps all alternatives and searches the plane for a globally consistent, slope-bounded curve. Explain the difference using the (s, x) maps and the singularity analysis.

Expected Output

- A validated DH model, home configuration and seven screw axes for the P6R whole body, with the assembled URDF.
- Numerical agreement between URDF, DH and PoE forward kinematics.
- The complete 6×7 screw Jacobian, the reduced fixed-x 6×6 Jacobian, ${}^{3}J_{5}$, and a numerical demonstration of which arm singularities remain singularities of the whole body.
- A complete analytical fixed-x IK solver with explicit branch labels, periodic-angle and joint-limit handling, and round-trip FK verification.
- Visualisations of all IK solutions for one representative tool pose.
- Branch-resolved reachable-position maps for two flange orientations and $x = -1, 0, +1$ m, with the four failure classes separated, and their world-frame overlay.
- Per-branch continuation plots for the fixed-x pick-and-place path, showing completion or the precise failure location.
- The (s, x) redundancy-plane maps for every branch, with the planned base trajectory $x(s)$ drawn on them.
- A comparison between greedy continuation and the branch-aware planner, with joint trajectories, joint-limit margins, Cartesian residuals and singularity margins for the representative paths.

Useful resources

1 Course simulations and assets: https://epfl-lasa.github.io/eng-654/

2 UR12e description: Universal_Robots_ROS2_Description (ur_description package).

3 AgileX UMR dimensions: AgileX Robotics product documentation.